

Pakistan E-commerce Analytics Platform

End-to-End Data Engineering & Business Intelligence Portfolio

Raw Data -> ETL -> PostgreSQL -> dbt -> Analytics -> BI Dashboard

October 07, 2026

Currency: PKR | Geography: Pakistan | Data: Synthetic

Table of Contents

- 1. Project Overview
- 2. Architecture & Technology Stack
- 3. Dataset Statistics
- 4. Data Pipeline & Quality
- 5. Data Warehouse Design
- 6. Analytics KPIs & Business Insights
- 7. Dashboards & Deliverables
- 8. How to Run
- 9. Interview Talking Points

1. Project Overview

This portfolio project demonstrates a production-style analytics platform for a fictional Pakistani e-commerce marketplace. It covers the full data lifecycle from synthetic data generation through ETL, validation, PostgreSQL warehousing, dbt transformations, data quality testing, analytics marts, BI dashboards, and business insights.

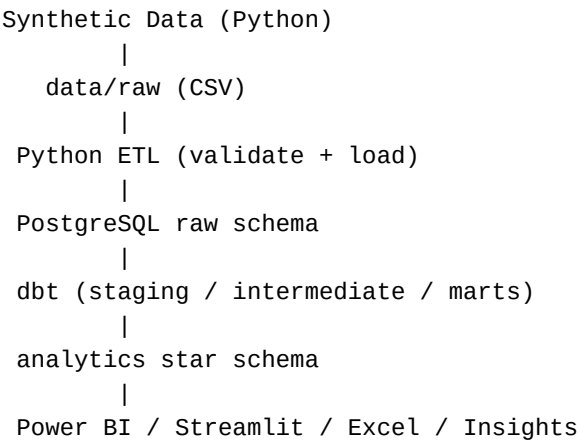
IMPORTANT: This project uses synthetic data generated to simulate a Pakistani e-commerce marketplace. It is intended for demonstrating data engineering and analytics skills and does not represent actual company data.

Business Questions Answered

- Revenue trends, AOV, and category/province performance
- Customer segments, RFM analysis, and lifetime value
- Seller performance, return rates, and cancellation rates
- Delivery times, on-time rates, and logistics by city
- Payment method usage and success rates
- Return reasons and refund impact

2. Architecture & Technology Stack

Pipeline Architecture



Technology Stack

Layer	Technology
Language	Python 3.11+
Database	PostgreSQL 16
Transformations	dbt Core + dbt-postgres
Orchestration	Apache Airflow
Validation	Python + pytest + dbt tests
BI	Power BI + Streamlit dashboard
Infrastructure	Docker Compose

3. Dataset Statistics

Entity	Rows
Locations	28
Categories	15
Sellers	1,000
Customers	100,000
Products	5,000
Orders	300,000
Order Items	901,205
Payments	300,000
Shipments	300,000
Returns	30,000
Total Raw Load	1,937,248

Date range: 2023-01-01 to 2025-12-31 | Currency: PKR | Random seed: 42

Pakistan-specific data includes major cities (Karachi, Lahore, Islamabad, etc.), provinces, PKR pricing by category, Cash on Delivery payment mix, and local carriers (TCS, Leopards).

4. Data Pipeline & Quality

ETL Stages

- Generate: src/data_generation/generate.py
- Validate: src/validation/validate_raw.py
- Load: src/ingestion/pipeline.py (PostgreSQL COPY)
- Transform: dbt run (30 models)
- Test: dbt test (64 tests) + pytest (8 tests)
- Export: Excel summaries + business insights markdown

Quality Checks

- Completeness: required IDs not null
- Uniqueness: primary keys unique
- Referential integrity: FK validation
- Accepted values: order status, payment method, return reasons
- Business rules: quantity > 0, delivery >= shipment date

Test Suite	Result
dbt test	64 / 64 passed
pytest	8 / 8 passed
Raw validation	All tables passed

5. Data Warehouse Design

Schemas

- raw - landing zone
- staging - dbt cleaned views
- intermediate - business logic
- analytics - star schema marts
- ops - pipeline monitoring

Star Schema Dimensions

dim_customer, dim_product, dim_seller, dim_category, dim_location, dim_date, dim_payment_method, dim_order_status

Fact Tables (with grain)

Fact Table	Grain
fact_orders	One row per order
fact_order_items	One row per order item
fact_payments	One row per payment
fact_shipments	One row per shipment
fact_returns	One row per return

KPI Definitions

- Gross Revenue = $\text{SUM}(\text{quantity} * \text{unit_price})$
- Net Revenue = Gross - discounts - refunds
- AOV = Total Revenue / Total Orders
- Return Rate = Returned Orders / Total Orders
- CLV = Historical net revenue per customer (portfolio proxy)

6. Analytics KPIs & Business Insights

KPI	Value
Total Orders	300,000
Customers with Orders	59,807
Gross Revenue	PKR 121,793,639,155.27
Net Revenue	PKR 113,376,438,051.32
AOV	PKR 387,810.60
Cancellation Rate	7.09%
Return Rate	10.00%

Top Cities by Revenue

City	Province	Revenue (PKR)	Orders
Karachi	Sindh	35,835,299,357	87,802

City	Province	Revenue (PKR)	Orders
Lahore	Punjab	28,964,463,204	71,409
Faisalabad	Punjab	7,824,022,105	19,188
Rawalpindi	Punjab	5,678,033,624	14,042
Multan	Punjab	4,768,836,439	11,825
Gujranwala	Punjab	4,621,753,148	11,352
Peshawar	Khyber Pakhtunkhwa	4,357,192,936	10,822
Islamabad	Islamabad Capital Territory	4,112,351,853	10,297

Top Categories by Revenue

Category	Revenue (PKR)
Mobile Phones	56,756,936,036
Computers & Laptops	29,165,411,804
Furniture	11,062,438,720
Home Appliances	10,655,776,202
Electronics	6,777,575,556
Fashion	1,825,467,319
Home & Kitchen	1,374,446,938
Sports	1,124,801,254

Payment Method Distribution

Payment Method	Transactions
Cash on Delivery	134,994
Mobile Wallet	59,980
Credit Card	45,076
Debit Card	36,002
Bank Transfer	23,948

Key Findings

Insights were generated from live SQL queries against the analytics schema, not fabricated.

- Revenue concentrates in Karachi, Lahore, and Faisalabad.
- Mobile Phones is the highest-revenue category.
- Cash on Delivery dominates payment transactions (realistic for Pakistan).
- Return rates vary by category; high-revenue categories warrant quality review.
- Monthly net revenue shows strong upward trend from 2023 to 2025.
- Repeat customers contribute a significant share of total spend.

7. Dashboards & Deliverables

Deliverable	Location
Streamlit BI Dashboard	dashboards/streamlit/app.py (localhost:8
Power BI Project	dashboards/powerbi/PakistanEcommerce/Pak
Excel Summaries	data/exports/analytics_summaries.xlsx
Business Insights	analysis/business_insights.md
DAX Measures	dashboards/powerbi/dax_measures.md
Data Dictionary	docs/data_dictionary.md
Interview Guide	docs/interview_guide.md

Power BI Dashboard Pages

- Page 1: Executive Overview - KPIs, revenue trend, category/province charts
- Page 2: Sales Analytics - monthly revenue, top products
- Page 3: Customer Analytics - segments, RFM, CLV
- Page 4: Product & Seller Analytics - performance tiers
- Page 5: Logistics - delivery times, on-time rate
- Page 6: Returns & Payments - return reasons, payment success

8. How to Run

1. `cp .env.example .env`
2. `pip install -r requirements.txt`
3. `docker compose up -d postgres`
(or: `python scripts/start_embedded_postgres.py`)
4. `python -m src.data_generation.generate`
5. `python -m src.validation.validate_raw`
6. `python -m src.ingestion.pipeline`
7. `cd dbt && dbt run && dbt test`
8. `streamlit run dashboards/streamlit/app.py`
9. Open `dashboards/powerbi/PakistanEcommerce/PakistanEcommerce.pbip`

9. Interview Talking Points

- Why PostgreSQL? Open-source, strong SQL, dbt support, Docker-friendly.
- Why star schema? Clear grains, BI-friendly, reusable dimensions.
- Why dbt? Version-controlled SQL, tests, documentation, lineage.
- Why Airflow? Orchestrates multi-stage pipeline with dependencies.
- How handle data quality? Python pre-load checks + dbt tests + pytest.
- Grain of fact_order_items? One row per order line item.
- How scale to 10M orders? Partitioning, incremental dbt, cloud warehouse.

Generated automatically by `scripts/generate_project_pdf.py` from project documentation and live warehouse queries. Regenerate after pipeline runs to refresh KPIs.